

Importance of Cryptography: HTTP vs HTTPS Credential Transmission

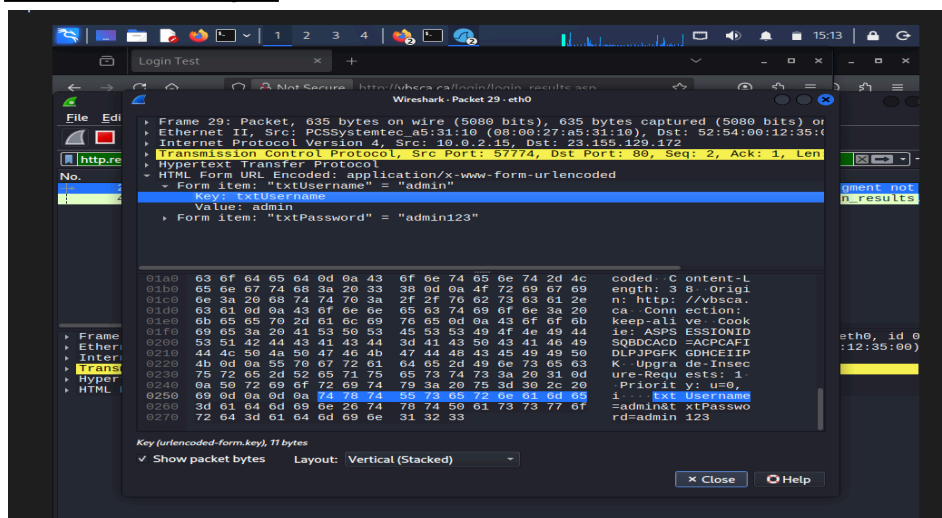
Objective

The main objective of this practical was to understand how login credentials are transmitted over HTTP and HTTPS. Wireshark was used to capture the network traffic and observe the difference between an unencrypted HTTP connection and an encrypted HTTPS connection.

Procedure

For this experiment, dummy login credentials were entered on an HTTP-based test login page and an HTTPS-secured page. At the same time, Wireshark was used to capture the network traffic through the eth0 interface. The captured packets were then examined to see whether the username and password could be viewed directly.

1. HTTP Traffic Analysis



Observation:

Packet 29 shows an HTTP POST request being sent to vbsca.ca/login/login_results.asp through port 80. When the packet was inspected in Wireshark, the submitted form data could be seen clearly.

The captured information included:

- txtUsername = admin
- txtPassword = admin123

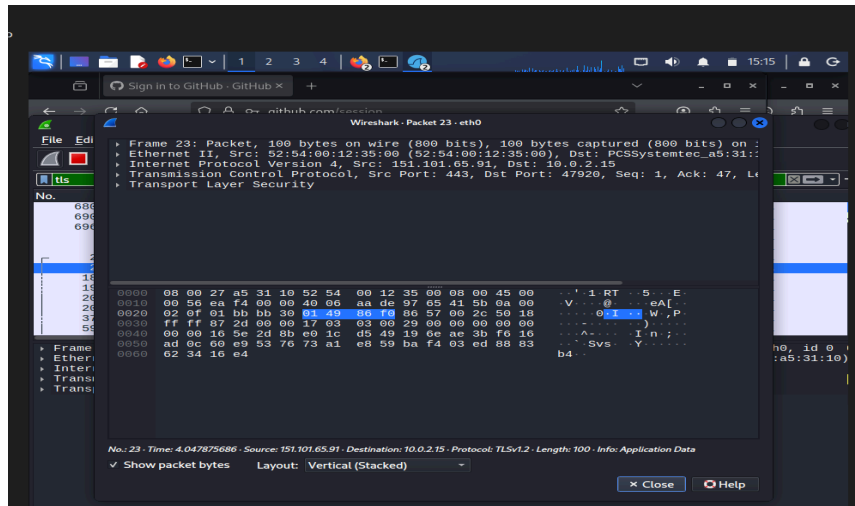
The same information could also be seen as readable text in the packet's raw data section:

txtUsername=admin&txtPassword=admin123

Explanation: This shows that HTTP does not encrypt the information being transmitted. Since the username and password are sent as plain text, someone who is able to capture the network traffic could potentially read the credentials directly.

This is why using HTTP for login pages is unsafe, especially when sensitive information such as passwords or personal details is being transmitted.

2. HTTPS Traffic Analysis



Observation:

Packet 23 shows TLS v1.2 Application Data being transmitted through port 443. Unlike the HTTP packet, the username and password cannot be seen directly in Wireshark.

The packet contains encrypted data that appears as unreadable characters in the hex/ASCII section. Information such as the source IP, destination IP, protocol, and packet length can still be observed, but the actual contents of the communication are protected.

Explanation:

This happens because HTTPS uses TLS encryption to protect the data before it is sent across the network. Even if someone captures the packets using a tool such as Wireshark, they cannot simply read the username and password from the captured traffic.

Therefore, HTTPS provides much better protection for sensitive information during transmission.

3. Comparison

Aspect	HTTP	HTTPS
Data format	Plaintext	Encrypted using TLS
Port	80	443
Risk of credential interception	high	Much lower
confidentiality	Not provided	Provided through encryption
integrity	Not provided by HTTP itself	Protected by TLS

4. Role of Encryption

This experiment helped demonstrate the importance of encryption when sending sensitive information over a network.

With HTTP, the login credentials are transmitted as plain text. If someone is monitoring the network traffic, they may be able to see the username and password without needing to decrypt anything.

HTTPS solves this problem by using TLS encryption. The data is encrypted before it travels across the network, so even if someone captures the packets, the actual contents cannot be easily understood.

TLS also provides other important security benefits, such as integrity, which helps detect whether data has been modified during transmission, and authentication, which helps verify that the user is communicating with the intended server

5. Conclusion

The Wireshark experiment clearly showed the difference between HTTP and HTTPS. In the HTTP capture, the dummy username and password were visible in plain text, showing how easily credentials can be exposed when encryption is not used.

In the HTTPS capture, the same type of sensitive information was protected by TLS and could not be directly read from the captured packets.

Therefore, cryptography plays an important role in protecting sensitive information while it is being transmitted over a network. HTTPS should be used instead of HTTP whenever login credentials, personal information, payment details, or other sensitive data are being exchanged.